Total No. of Questions : 8] [Total No. of Printed Pages : 3

Paper Code : 21309

F-409

B.C.A. (Second Semester) Examination, 2023

(New Course)

Paper No. BCA-N-204

PROGRAMMING IN 'C'

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks.

1. (a) What are "reserved-words" in C-programming language?

   (b) Describe various logical operations in C-language with proper example of each.

2. (a) How 'while' loop is different from 'do-while'? Write syntax for each.

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   (b) Write a program using 'C' language to find factorial of any integer number.

3. (a) How an array is different from a structure? Write syntax of both.

   (b) Using 'recursion' write any one program in C-programming language.

4. (a) Differential 'call-by-value" and "call-by-reference" in detail.

   (b) Write one program in 'C' to demonstrate the concept of string manipulation by string handling function.

5. (a) What is pointer? How it is implemented? Explain.

   (b) How "MALLOC" is different from "CALLOC"? (Malloc ( ) vs. calloc ( )).

6. (a) Write one program in C for showing creation & deletion of file.

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(b) Is structure & union same things? (Yes/ No). Justify your answer with one small supporting code.

7. (a) What are standard 'C' library function? Describe each.

(b) Describe resting of loops with suitable code in C-language.

8. Write in brief about : (any four) of the following:

(i) Functions in C-language

(ii) Program for prime number testing

(iii) Relational operators

(iv) Preprocessor directive

(v) Role of getch () function

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Paper Code : 21309

F–409

B. C. A. (Second Semester) EXAMINATION, 2019

(New Course)

Paper No. BCA—N—204

PROGRAMMING IN 'C'

Time : Three Hours] [*Maximum Marks : 70*

Note : Attempt any *five* questions. All questions carry equal marks.

1. (a) What is a data type ? Explain various data types available in 'C' language with syntax and examples. 7

   (b) What are various categories of operations available in 'C' language ? Describe any *two* category operations. 7

2. (a) What is an array ? How many types of arrays are available in 'C' language ? Explain with proper example. 7

   (b) Write a program in 'C' language to display tables from 2 to 20. 7

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3. (a) Distinguish between procedural and modular structed programming. Explain each with examples. 7

   (b) What is recursion ? Write a program to calculate factorial of a number using recursion. 7

4. (a) How are pointer and arrays related ? Describe malloc( ) and calloc( ) using pointer. 7

   (b) How call-by-reference can be implemented using pointers ? Write a program to swap values of two variables using call-by-reference. 7

5. (a) What is a file ? Write various attributes and operations related to files. 7

   (b) What is a macro ? Describe its definition and calling using file concept. 7

6. (a) What is the use of storage classes in 'C' language ? Describe various storage classes with proper examples. 7

   (b) That is the use of loop in 'C' language ? Write various available loops in 'C' language and explain their use with examples. 7

7. (a) What is difference between switch and if-else statements ? Write a program in 'C' language using switch statement. 7

   (b) Taking suitable examples differentiate between break and continue statement. 7

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8. Write shorts Notes on any *four* of the following : $3\frac{1}{2}$ each

(a) Comma operator

(b) Dynamic memory allocation

(c) File appending and deleting

(d) Nested structure

(e) Union data structure

(f) Parameter passing methods

(g) Pointer arithmetic.

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Paper Code : 21309
F–409

B. C. A. (Second Semester) EXAMINATION, 2018

(New Course)

Paper No. BCA—N—204

PROGRAMMING IN C

Time : Three Hours] [*Maximum Marks : 70*

Note : Attempt any *five* questions. All questions carry equal marks.

1. (a) List the various types of operators available in 'C'. Write the hierarchical order of evaluation of these. 7

   (b) Write a function which accepts an array of size *n* containing integer values and returns the average of all values. Call the function from main program. 7

2. (a) What is an array ? Show different methods to declare and initialize one-dimensional array. Draw how an integer array of 6 elements having base address 5001 is stored in memory. Explain actual storage of two dimensional array *d* [3] [4] in the memory with diagram. 7

   (b) Write and explain the action of 'while' statement. Develop a program in 'C' language to compute the average of every third integer number lying between 1 and 100. Include appropriate documentation. 2, 5

3. (a) Explain with the help of an example and flow-charts the action of break statement and the continue statement. 7

   (b) What are the disadvantages of vested if else statement ? Give an example of nesting of loops. 5, 2

4. (a) Write a 'C' program to find out sum of diagonal elements of a matrix. 7

   (b) Input two different arrays of integers. Write a 'C' program using pointer variables to merge them into third array of sufficient size. 7

5. (a) How does passing an array as an argument to a function differ from call by value ? Is it legal for a function f 1 and f 2, which then calls f 1 ? Justify your answer. 5, 2

   (b) Define recursion. Write a C program to find out factorial of a given number using function recursion. 2, 5

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6. (a) What is dynamic memory allocation ? Mention four functions used for dynamic memory manipulation. Give an example of dynamic memory allocation. 1, 4, 2

   (b) What do you mean by macros ? Write a macro that clears an array to zero. 3, 4

7. (a) Give the main advantages of storing data as a file. Describe various ways in which data files can be categorized in 'C'. Illustrate by examples. 1, 4, 2

   (b) What do you mean by a pointer variable ? Write a function in 'C', using pointers for the array of elements, for sorting the elements. 1, 6

8. (a) What are the various storage classes available in 'C' ? Explain and give *one* example of each. 7

   (b) Write a function to accept two strings as arguments. The function should return the index of the second storing in the first string, if it exists, and –1 otherwise. Don't use any library routines related to strings. 7

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